

Week 2 - Quiz

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Which of the following is the correct expression used to calculate the fracture toughness K_{Ic} of a material? In this equation c is the crack length, Y is a dimensionless constant that depends upon the loading configuration and σ_c is the fracture stress.

Select one:

- ☐ a. $K_{Ic} = Y\sigma_c\pi \cdot c = 1 \times \text{Nm}^{-2} \times 1 \times \text{m} = \text{Nm}^{-1}$ ✗
- ☐ b. $K_{Ic} = Y\sigma_c\sqrt{\pi \cdot c} = 1 \times \text{Nm}^{-2} \times \sqrt{1 \times \text{m}} = \text{Nm}^{-3/2}$ ✓
- ☐ c. $K_{Ic} = Y\sigma_c^2\sqrt{\pi \cdot c} = 1 \times (\text{Nm}^{-2})^2 \times \sqrt{1 \times \text{m}} = \text{N}^2\text{m}^{-7/2}$ ✗
- ☐ d. $K_{Ic} = Y\sigma_c^3\sqrt{\pi \cdot c} = 1 \times (\text{Nm}^{-2})^3 \times \sqrt{1 \times \text{m}} = \text{N}^3\text{m}^{-5/2}$ ✗

$$K_{Ic} = \text{MPa m}^{3/2} = \text{MN m}^{-3/2} \text{ (From Notes)}$$

Which of the following are the correct units for thermal conductivity?

Select one:

- ☐ a. W/K ✗
- ☐ b. Wm/K ✗
- ☐ c. W/(m²K) ✗
- ☐ d. W/(m K) ✓

$$q = -K \frac{dT}{dx}$$

$$\therefore K = \frac{q dx}{dT} = \frac{\left(\frac{\text{W}}{\text{m}^2}\right) \times \text{m}}{\text{K}} = \frac{\text{W}}{\text{m K}}$$

Which of the following materials groups contains materials that are the most difficult to recycle?

Select one:

- ☒ a. Polymer composites → Because fibres and polymer are bound in a matrix.
- ☐ b. Glass → V. hard to reuse and recycle fibres completely.
- ☐ c. Thermoplastic polymer → Can be melted and reused.
- ☐ d. Metal alloy →

Clear my choice

The phase lag, δ is a measure of vibrational energy dissipation. Which of the following statements describes it most accurately?

Select one:

- ☐ a. It is a measure of the energy absorbed during cyclic loading ✗
- ☐ b. It measures the work done on a test specimen during a fatigue test ✗
- ☐ c. It is a measure of the phase difference between the applied stress and the measured strain ✓
- ☐ d. It is a measure of the duration of the load cycle during a constant frequency cyclic test ✗

What is the definition of the strength of a material

Select one or more:

- ☐ a. The stress at which the material can no longer return to its original shape once the stress is removed ✓
- ☐ b. The maximum strain that a materials can sustain before it breaks ✗ (-)
- ☐ c. The maximum stress that a materials can sustain before it breaks ✗ (Ultimate Tensile strength)
- ☐ d. The strain at which the material can no longer return to its original shape ✗ (-)

The hardness of a metal is an approximate measure of what?

Select one:

- ☐ a. The yield strength ✓
- ☐ b. The modulus ✗ (For an elastomer)
- ☐ c. The fatigue resistance ✗
- ☐ d. The creep resistance ✗

Which of the following statements is the most correct description of the glass transition temperature, T_g , of a polymer?

Select one:

- ☐ a. T_g is the temperature above which a polymer exhibits glassy like behaviour. ✗
- ☐ b. T_g is the boiling temperature for the polymer. ✗
- ☐ c. T_g is the temperature at which the polymer crystals in a semi-crystalline polymer melt. ✗
- ☐ d. T_g is the temperature above which an amorphous polymer exhibits rubber-like behaviour. ✓

● To be considered an elastomer a polymer usually has the following four requirements:

- The temperature is above the glass transition temperature, T_g .